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Investigating and modeling PET methanolysis under supercritical conditions by response surface methodology approach

Qinli Liu, Ruosong Li, Tao Fang *

Department of Chemical Engineering, Xi'an Jiaotong University, Xi'an 710049, China

HIGHLIGHTS

- Interaction effects were investigated in PET supercritical methanolysis process.
- RSM was used to develop mathematical model and analysis for process.
- A modified square root transformation model was used in RSM for better prediction.
- Terms including MeOH to PET mass ratio are uninfluential among experiment conditions.
- The optimum DMT yield was much closer to the extent of the complete reaction.

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ABSTRACT

Polyethylene terephthalate (PET) is one of the most widely used raw materials in chemical industry. Chemical recycling of the waste PET is a sustainable way not only considering the economic benefit but also the environmental goals. This paper focused on the effects among three foremost variables (methanolysis temperature, time, and mass ratio of methanol (MeOH) to PET), established a mathematical model depicted the complete effects of the parameters on the process, and sought the optimal solution of this PET methanolysis reaction. Response surface methodology (RSM) was used to model and analysis the recycling process of dimethyl terephthalate (DMT) from waste PET under supercritical methanol conditions. Methanolysis temperature and time could increase the DMT yield. Mass ratio of MeOH to PET hardly affected DMT yield when the methanol is excess. Different operation parameters including reaction temperature (240–320 °C), reaction time (15–120 min), and MeOH to PET mass ratio (6:1–10:1) were modeled and optimized using RSM. The quadratic equation with mathematical transformation (square root) modified was established for predicting the DMT yield and evaluated the validity on the basic of analysis of variance (ANOVA). The result indicated that the MeOH to PET mass ratio and relative term exhibit a less significant influence on the response surface in comparison to reaction temperature and time in the range of the experiment conditions. Prediction made by the developed model was in reasonably good agreement with the test runs. Under the optimal condition, the yield of DMT reached 99.79% which was much closer to the extent of the complete reaction.

1. Introduction

Polyethylene terephthalate (PET) is a long-chain polymer that falls into the category of the generic family of polyesters, synthesized using intermediates, terephthalic acid (TPA) and ethylene glycol (EG). As a thermoplastic polymer resin of the polyester family, it is widely used in producing synthetic fibers, beverage or other containers, engineering resins and assisting thermoforming applications [1]. The Pira Company released the prediction that the consumption of PET only in global packaging industry would grow to over 20 million tons by 2019 [2]. Therefore, it is one of the most important raw materials in chemical industry. Considering the current energy crisis, PET, as one of the products of petroleum, is also facing up with the challenge from the limitation of the fossil resource. What's more, with the increasing consumption of PET, the urgent necessity for recycling and reprocessing the waste PET has arisen. Actually, it is a good solution to prevent the resource shortage, and release the awareness and concern for environmental pollution. It should be pointed out that PET does not create a direct hazard to the environment. Due to its substantial volume fraction in the waste stream and

* Corresponding author. Tel.: +86 2982664375.
E-mail address: taofang@mail.xjtu.edu.cn (T. Fang).

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its high resistance to the atmospheric and biological agents in degradation, it is eventually identified as a noxious material.

Based on theoretical analysis and industry experiences, different ways have been considered to recycle the PET: primary recycling [3], mechanical recycling [4,5], chemical recycling [6–10], and even recycling for supplying energy [11]. According to the principles of sustainable development, the only one acceptable choice mentioned above is chemical recycling, since it does not only serve as a basic solution to the solid waste problem, but also contribute to the conservation of raw petrochemical products and energy. Several processes for PET depolymerization have been proposed such as hydrolysis under the existence of an alkali [12], methanolysis in liquid or vapor methanol [13], and glycolysis in liquid ethylene glycol [7]. However, these methods have obvious problems, such as slow reaction [14], purification of monomer [15], using and separating catalyst [16,17].

In recent years, supercritical fluids (SCFs) have been focused on dealing with above problems because of their special reactivity. The supercritical fluid has high density as liquid state, and a high kinetic energy as gas molecule. Consequently, the reaction rate in supercritical condition is expected to be higher than that under liquid state condition. Sako and coworkers [18] firstly studied the characteristics of the depolymerization of poly (ethylene terephthalate) in supercritical methanol using a high-pressure batch reactor. PET could be completely converted into monomer, dimethyl terephthalate (DMT) and EG, at 300 °C and 30 min without catalyst. Yang [19] reported that PET depolymerization in supercritical methanol were greatly affected by the temperature and reaction time. Goto and coworkers [20,21] used a batch reactor at temperatures of 300–350 °C under an estimated pressure of 20 MPa for a reaction time of 2–120 min. Furthermore, they investigated the reaction mechanism and kinetics of the depolymerization of PET to its monomers in supercritical methanol [22–25]. Although most researchers focused on the yield of products by single factor analysis with three significant parameters, reaction temperature, time and pressure or mass ratio of MeOH to PET (in an enclosed batch reactor, pressure is related with mass ratio of MeOH to PET which is much easier to be controlled), respectively [20,26–29], few studies have investigated the interaction effects among the process parameters so that it might lead to an inadequate or even inaccurate study for this process. Thus, the aim of this study is to consider the complete effects of the parameters on the process and discuss the interaction effects among the key variables (temperature, time and mass ratio of MeOH to PET) on depolymerizing waste PET using supercritical methanol technology. In order to achieve this target, a statistical way called response surface methodology (RSM) is utilized.

Response surface methodology (RSM), a progressive analysis method including a series of statistical and mathematical techniques, focuses on designing, improving and optimizing the system with several controllable variables. During RSM analysis, a mathematical model which describes the relationship between the response and independent variables is constructed to evaluate the significance of each variable and their interactions [9]. Compared with the RSM, single factor analysis is well known and widely used to determine the optimal system condition. The individual effect of the system variable is analyzed while keeping other conditions as constants. But the interactive effects among the system variables are neglected, and here comes the RSM to improve it. So far, RSM has a lot of applications in different kinds of areas, especially in chemical processes [30–33]. Also few researches have been done using RSM to discuss the recycling of PET, such as Katoch [34] used RSM to predict the optimal condition of glycolysis time and temperature in the recycling of PET.

In this work, the three-level-three-factor Box–Behnken design experiments based on RSM were conducted and predicted the response on the basis of the experimental factors by developing a suitable mathematical model. The complete effects of three parameters (reaction time, reaction temperature and mass ratio of MeOH to PET) on the yield of DMT were emphatic studied, evaluated and optimized. The data obtained will be utilized to further understand the PET supercritical methanolysis process.

2. Materials and methods

2.1. Materials

The raw materials used in the study were bottle-grade virgin PET granules ($d \approx 3$ mm) supplied by E.I. duPont de Nemours and Company (Japan). Methanol (99.5%, HPLC purity grade) was supplied by Jindong Tianzheng Chemical Reagent Co., Ltd. (P.R. China). Standard sample of DMT (>99.5%) was supplied by Dr. Ehrenstorfer Company (Germany).

2.2. Apparatus and procedures

The apparatus made by Huaxing Chemical Device Co., Ltd. (Nantong, P.R. China), consisting of three parallel stainless batch reactors (about 10 mL inner volume) and an electric heater, was used for the PET methanolysis. The apparatus has the special function of swinging the reactor to reduce the influence of mass transfer (swing span = 2 cm; frequency of swing = 2 Hz).

The three reactors charged PET and methanol were placed in the electric furnace at a desired temperature which was set in advance. During the process, the reactors were kept under the reaction conditions, then the reactors were cooled quickly to about 25 °C and the products were taken out. Water and methanol were used to carry out the whole mixed products. Cold water was added to dissolve the small quantity of EG absorbed in the surface of solid products and extract out DMT dissolved in methanol. The mixture was separated into solid and liquid phases by filtration. The volume of filtrate was accurately measured by graduated cylinder. About 1 mL filtrate was dissolved by the mixed solution (methanol/water = 3:1, volume ratio), and then analyzed with HPLC. The solid product that was mainly composed of DMT was then dried at 25 °C to a constant weight, and grounded to fine powders for HPLC analysis.

The HPLC (Beijing Puxi, L6-UV6) was equipped with an Alltima HP C18 column (4.6 mm × 250 mm), 5 μm, Alltech Associates Inc. The ultraviolet detector was set at 254 nm. Water (25%) in methanol (75%) was employed as a mobile phase flowing at 1.0 mL/min.

2.3. PET methanolysis estimation method

For the methanolysis of the ester bond between TPA and EG, the majority of PET was decomposed to the monomers DMT and EG in supercritical methanol, as shown in Fig. 1. Moreover, the amount of low molecular weight oligomer was also produced in the present reaction. The DMT in the solid products and the liquid products could completely be separated and detected accurately with HPLC method, respectively.

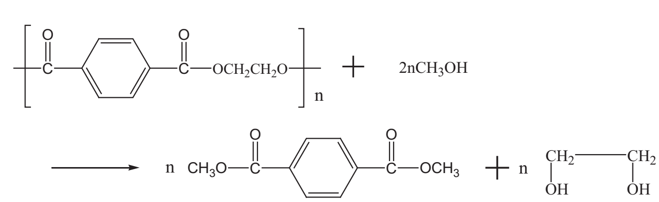

Fig. 1. Main reaction of PET depolymerization in supercritical methanol.

The yield of DMT is represented as the percentage of the DMT amount in the liquid and solid products, which is calculated from HPLC data against the theoretical DMT amount if the PET was completely depolymerized as Eq. (1):

$$\text{Yield of DMT} = \frac{\text{Weight of actual DMT in liquids products} + \text{Weight of actual DMT in solid products}}{\text{Theoretic output}} \quad (1)$$

2.4. Experimental design

In order to evaluate the effect of the reaction parameters and optimize the yield of DMT, a three-level-three-factor Box–Behnken design was applied on the basis of Design Expert software (version 8.0.6, Statease Inc., Minneapolis, USA). The Box–Behnken design which is one of response surface designs requires three-levels and can be a more efficient alternative to the full three-level factorial design. The Box–Behnken design involves three blocks with 4 factorial points, and three replications of the center points. The center points are used to measure the experimental errors and estimate the lack of fit. In this work, the experimental conditions (reaction temperature, time, and mass ratio of MeOH to PET) are three independent variables. In order to obtain a suitable parametric range of the reaction conditions for RSM, a series of experiments should be first conducted for single factor analysis. The depolymerization reactions were carried out in a large range at temperatures (220–320 °C), reaction time (15–120 min), and MeOH to PET mass ratio (4–10). Furthermore, the response design experiments were tested with three parallel reactors for three factors: reaction temperature (A), reaction time (B) and mass ratio of MeOH to PET (C), in ranges of 240–320(°C), 15–120 min and 6–10, respectively. The coded and uncoded independent variables and levels used in the experimental design are shown in Table 1. There are a total of 15 three-level-three-factor Box–Behnken design experiments and responses as shown in Table 2, which are completely randomized to eliminate system errors. The central point was replicated three times to consider the reproducibility.

Table 1
Coded and uncoded levels of variables for RSM Box–Behnken design.

Variable	Symbol	Coded factor levels		
		−1	0	1
Reaction temperature (°C)	A	240	280	320
Reaction time (min)	B	15	68	120
MeOH to PET mass ratio	C	6	8	10

Table 2
Experimental design based on RSM for the yield of DMT.

Run order	Temperature (°C) X_1	Time (min) X_2	MeOH/PET mass ratio X_3	Yield (%)
1	280	68	8	76.34
2	280	15	10	21.18
3	280	120	10	93.39
4	320	68	6	90.96
5	240	68	6	52.41
6	320	68	10	96.57
7	280	68	8	78.35
8	280	68	8	79.37
9	240	15	8	6.73
10	240	68	10	52.31
11	240	120	8	76.16
12	280	15	6	21.03
13	280	120	6	91.46
14	320	120	8	99.79
15	320	15	8	28.11

3. Results and discussion

3.1. Single factor analysis of experimental factors

To study the influence of the reaction temperature on the yield of DMT, the experiment was conducted at 220, 240, 260, 280, 300 and 320 °C, respectively. Fig. 2 shows the plots of the yield of DMT versus reaction temperature. The yield of the monomer increased sharply with the methanol critical temperature. However, above

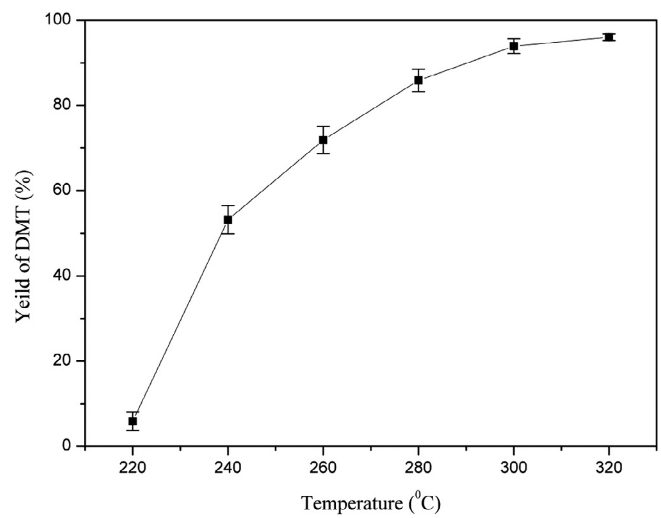

Fig. 2. Relationship between the yield of DMT and the reaction temperature on depolymerization of PET in supercritical methanol. Reaction time = 60 min, PET/MeOH = 1/10 (mass base), swing span = 2 cm, and frequency of swing = 2 Hz.

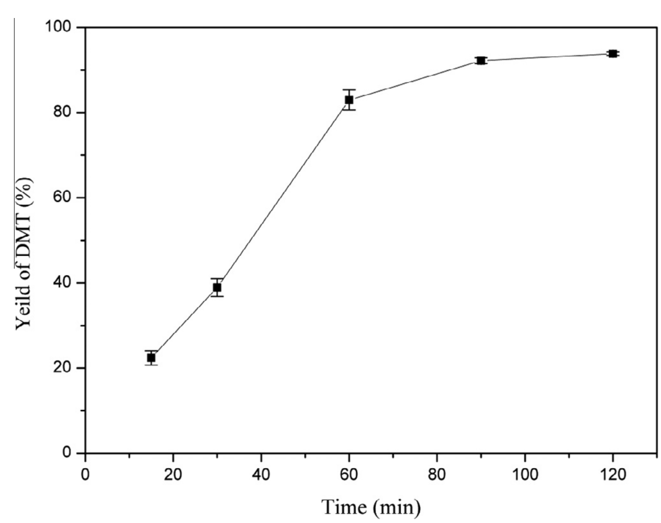

Fig. 3. Relationship between the yield of DMT and the reaction time on depolymerization of PET in supercritical methanol. Reaction temperature = 280 °C, PET/MeOH = 1/10 (mass base), swing span = 2 cm, and frequency of swing = 2 Hz.

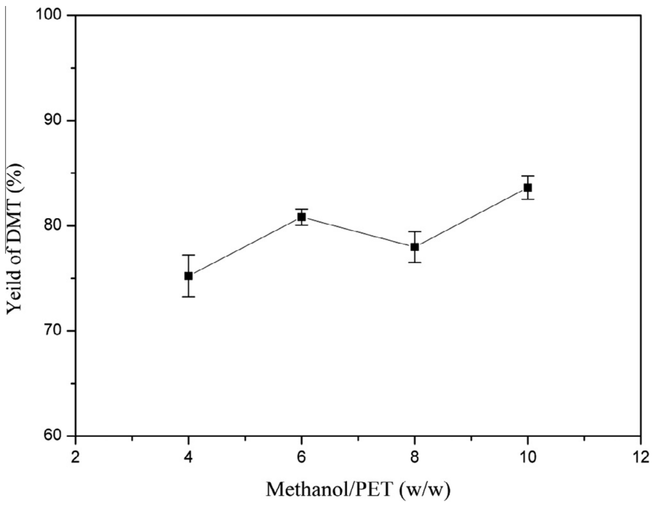

Fig. 4. Relationship between the yield of DMT and mass of MeOH to PET on depolymerization of PET in supercritical methanol. Reaction temperature = 280 °C, time = 60 min, swing span = 2 cm, and frequency of swing = 2 Hz.

the critical temperature, the increasing temperature was relatively rarely attributed to the higher yield of DMT.

The effect of the reaction time on the depolymerization is depicted in Fig. 3. The experiments were carried out for separately 15, 30, 60, 90, and 120 min (including the time required for reaching the prescribed condition). The yield of DMT was notably enhanced as the time goes on. The yield of DMT respectively reached 92.18% and 93.83% with the reaction times of 90 and 120 min at 280 °C. The yield of DMT approached a constant value which closes to the extent of complete depolymerization.

To investigate the effect of mass ratio of MeOH to PET on depolymerization at 280 °C in 60 min reaction time, the experiments were conducted and the data are summarized in Fig. 4. The result indicates that the yield of DMT increases slowly when the mass ratio is between 4 and 10. There was a slightly increase between 6 and 10. So 6–10 as the RSM experiment condition were chosen.

3.2. Determination of the regression model

The model used in RSM is generally a quadratic equation for predicting response as a function of independent variables. The general form of this quadratic polynomial can be expressed as:

Table 4
Evaluate the five mathematical transformed models.

Transformation	Equation	Model P-value	Lack of fit P-value	Adjusted R-squared	Predicted R-squared
None	$y' = y$	0.0004	0.0382	0.9628	0.7923
Square root	$y' = \sqrt{y}$	<0.0001	0.1115	0.9927	0.9609
Natural log	$y' = \ln y$	0.0003	0.0112	0.9659	0.8063
Inverse	$y' = \frac{1}{y}$	0.0712	<0.0001	0.6572	−0.9588
Power	$y' = y^2$	0.0050	0.0249	0.8914	0.3884

$$Y = \alpha_0 + \sum_{i=1}^{3}\alpha_i X_i + \sum_{i=1}^{3}\alpha_{ii} X_i^2 + \sum_{i=1}^{2}\sum_{j=i+1}^{3}\alpha_{ij} X_i X_j \tag{2}$$

where Y is the predicted response, α_0 is a constant, α_i, α_{ii} and α_{ij} are linear, quadratic and interactive coefficients, respectively. X_i and X_j indicate levels of the input variables (reaction temperature, time, and mass ratio of MeOH to PET). The coefficients of the polynomial were estimated by least square fitting method. Validation of the model was investigated by analysis of variance (ANOVA).

The previous experimental results were indicated the suitable range of experiment conditions for RSM in the above single factor analysis. The design matrix in actual terms and the experimental results of RSM are presented in Table 2. The experimental data were fitted to Eq. (2) shown as Eq. (3), resulting in a mathematical regression quadratic model to explain the relationship between the yield of DMT and the experimental factors:

$$\begin{aligned}\text{DMT yield} = {} & 0.78 + 0.16X_1 + 0.35X_2 + 0.0095X_3 \\ & + 0.0056X_1X_2 + 0.014X_1X_3 + 0.0045X_2X_3 \\ & - 0.045X_1^2 - 0.21X_2^2 - 0.0044X_3^2\end{aligned} \tag{3}$$

ANOVA for studying the significance of fit from quadratic equation for the experimental data is shown in Table 3. The P-values lower than 0.05 indicated that model terms were significant. Although the P-value of the regression equation is much less than 0.05, the lack-of-fit test for the model is significant with a P-value of 0.0382. So, the model is unsuitable for fitting the experimental data.

In order to reasonable fit the real experiment data, mathematical transformation should be used to modify the quadratic model. Five common mathematical transformations were tried to modify the model. Four important evaluate indexes for the new models were summarized in Table 4. The square root transformation is

Table 3
ANOVA for the quadratic model.

Source	SS[a]	DF[b]	MS[c]	F	P	Significant[d]
Model	1.38	9	0.15	41.30	0.0004	Yes
Temperature (X_1)	0.20	1	0.20	55.16	0.0007	Yes
Time (X_2)	1.01	1	1.01	271.85	<0.0001	Yes
MeOH/PET mass ratio (X_3)	7.207E-4	1	7.207E-4	0.19	0.6775	No
X_{12}	1.272E-4	1	1.272E-4	0.034	0.8602	No
X_{13}	8.185E-4	1	8.185E-4	0.22	0.6580	No
X_{23}	7.922E-5	1	7.922E-5	0.021	0.8894	No
X_1^2	7.525E-3	1	7.525E-3	2.03	0.2133	No
X_2^2	0.16	1	0.16	43.19	0.0012	Yes
X_3^2	7.300E-5	1	7.300E-5	0.020	0.8938	No
Residual	0.019	5	3.702E-3			
Lack of fit	0.018	3	6.012E-3	25.36	0.0382	Yes
Pure error	4.741E-4	2	2.370E-4			
Total	1.39	14				

[a] Sum of squares.
[b] Degrees of freedom.
[c] Mean square.
[d] $P < 0.05$ were considered to be significant.

Table 5
ANOVA for the improved model by square root transformation.

Source	SS[a]	DF[b]	MS[c]	F	P	Significant[d]
Model	0.77	9	0.085	212.08	<0.0001	Yes
Temperature (X_1)	0.098	1	0.098	244.19	<0.0001	Yes
Time (X_2)	0.54	1	0.54	1353.01	<0.0001	Yes
MeOH/PET mass ratio (X_3)	1.995E-4	1	1.995E-4	0.50	0.5125	No
X_{12}	5.214E-3	1	5.214E-3	12.97	0.0155	Yes
X_{13}	2.210E-4	1	2.210E-4	0.55	0.4918	No
X_{23}	1.768E-5	1	1.768E-5	0.044	0.8422	No
X_1^2	6.271E-3	1	6.271E-3	15.60	0.0109	Yes
X_2^2	0.12	1	0.12	286.90	<0.0001	Yes
X_3^2	5.478E-5	1	5.478E-5	0.14	0.7271	No
Residual	2.010E-3	5	4.020E-4			
Lack of fit	1.857E-3	3	6.191E-4	8.13	0.1115	No
Pure error	1.524E-4	2	7.619E-5			
Total	0.77	14				

[a] Sum of squares.
[b] Degrees of freedom.
[c] Mean square.
[d] $P < 0.05$ were considered to be significant.

the best choice for this process. The equation of the model is shown in Eq. (4), in which y presents the DMT yield. ANOVA for the improved model is presented in Table 5:

$$\sqrt{y} = 0.88 + 0.11X_1 + 0.26X_2 + 0.0050X_3 - 0.036X_1X_2 + 0.0074X_1X_3 + 0.0021X_2X_3 - 0.041X_1^2 - 0.18X_2^2 + 0.0039X_3^2 \quad (4)$$

The ANOVA (Table 5) shows that the improved quadratics model is the most significant for the P-value is much lower than 0.05. And the lack of fit test for the improved model is not significant with the P-value of 0.1115. Moreover, the determination coefficient ($R^2 = 0.9974$) shows that only 0.26% of the total variation could not be illuminated from the regression model. And the means of the adjusted coefficient of determination ($R^2_{adj} = 0.9927$) also corroborates that the modified model estimates the experimental data with high-degree of fit. The value R^2_{adj} is close to the determination coefficient value (R^2). All above indicate that the regression quadratics model with square root transformation could fit the experimental data very well. This model can be used to predict the DMT yield for any combination of experimental conditions within the range of experimental factors. The models developed using response surface methodology cannot be used for experimental conditions outside the range of experimental parameters.

The significance of each coefficient determined by P-values was illustrated in Table 5. The principal effect (A, B), the quadratic (A^2, B^2) and the first-order interaction effect (AB) were the significant terms in the supercritical methanolysis process for PET. Therefore, omitting the non-significant terms (C, AC, BC, and C^2), the final simplified model is presented as follow:

$$\sqrt{y} = 0.89 + 0.11X_1 + 0.26X_2 - 0.036X_1X_2 - 0.041X_1^2 - 0.18X_2^2 \quad (5)$$

3.3. Three-dimensional (3D) response surface and contour plots

Three-dimensional response surface plots are a function of two factors with the other factor at constant, presented as a smooth surface framed in the three-dimensional space. These graphics help to understand the main and interaction effects of the factors. In this study, one variable was as a constant at the center level when the other variables are changing. The 3D response surface plots and the corresponding counter plots of the DMT yield versus interactions of two variables are illustrated in Fig. 5.

The interaction effect between temperature and time on the DMT yield is depicted in Fig. 5a. With the increment of temperature and depolymerization time, the yield of DMT generally shows a rising trend. With the increases in temperature and depolymerization time, the yield of DMT generally shows an increasing trend. Furthermore, it is observed that the DMT yield increased with temperature at any fixed temperature under given conditions. It approaches a nearly steady state at about 320 °C. This is consistent with our previous single factor conclusion. Both analysis results indicate that the methanol have a higher capacity for dissolving PET under the supercritical state than that in normal liquid methanol. The ester in PET could be well dispersed in the supercritical methanol, which facilitates the depolymerization. Zhu [35] thought that temperature rise may increase or decrease the solubility of the solid depending on the system in the supercritical fluids. In this system, the growing speed of PET dissolution in supercritical methanol was going down with increasing the temperature. It can be explained that the solubility decline is becoming more significant due to a rise in temperature.

When keeping the temperature as a constant, DMT yield is increased with the time under given conditions agree with our previous single factor analysis for time. This result could be explained by report [22] that the PET degradation in the supercritical methanol were consecutive reactions. The initial stage of PET depolymerization in supercritical methanol was the random scission in the heterogeneous phase and the final stage was the specific scission in the homogeneous phase. The final reaction was the rate-determining step and methyl-(2-hydroxyethyl) terephthalate (MHET) was a persistent intermediate product. At the beginning of the reaction, the first reaction proceeded predominantly in a faster speed. When most PET turned to the intermediate product, the concentration of the MHET decreasing resulted in a mild increase of the DMT yield.

The interaction of MeOH to PET mass ratio with temperature and time are depicted in Fig. 5b and c, respectively. The two figures depict that MeOH to PET mass ratio has little effect on the yield of DMT. The results could also be obtained from the AVONA of the response surface model. Therefore, the MeOH to PET mass ratio and relative term exhibit a less significant influence on the response surface in comparison to reaction temperature and time. One reason could be that the amount of methanol for completely reacting with 0.4 g PET was calculated using the material conservation and only 0.13 g methanol should be used for reaction. In the batch

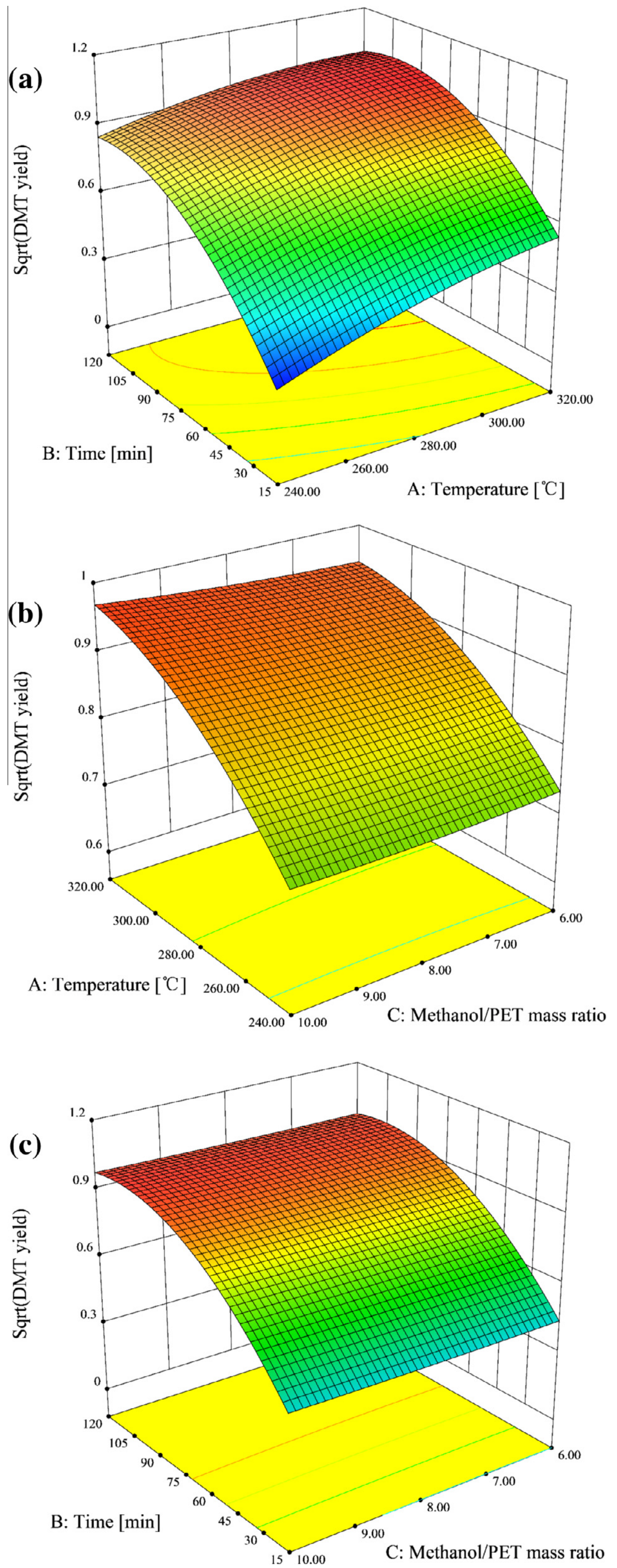

Fig. 5. Response surface plot and contour plot. (a) Reaction temperature and time at MeOH/PET mass ratio of 8; (b) reaction temperature and MeOH/PET mass ratio at time of 68 min; (c) time and MeOH/PET mass ratio at 280 °C.

reactor, the more mass of methanol should be used, the higher pressure will be achieved. The excess of methanol, 1.6–4 g, could ensure the complete reaction and achieve a desired pressure. When the amount of methanol related to pressure in the sealed batch reactor, the previous thermodynamic research studied the properties in high pressure could use to explain the phenomenon [36]. The reaction in the supercritical state had a big negative partial molar volume changing near the critical pressure. However, when pressure is further increasing, the change of partial molar volume tended to zero, and the reaction pressure had small influence on the reaction. Therefore, guaranteeing a faraway critical pressure, the mass ratio of MeOH to PET had small effect on the reaction.

Table 6
Comparisons of DMT yield obtained with different reaction condition.

References	Optimum reaction conditions			DMT yield (%)
	Reaction temperature (°C)	Reaction time (min)	MeOH to PET mass ratio	
This study	298	112	6	99.79
Kim [28]	310	60	10	97.7
Sako [18]	300	30	15 mL MeOH to 0.5 g PET	~100
Yang [19]	260–270	40–60	6–8	~98
Peng [37]	330	45	43	73

3.4. Optimization of PET methanolysis by RSM

The process conditions for the PET depolymerization were optimized with the aim of getting the maximum DMT yield. The optimization of variables in this study was carried out using the same software of RSM. The independent parameters which include reaction temperature, reaction time and mass ratio were set within the range between low (−1) and high (+1) as shown in Table 1. The optimum operating conditions under the experiment range suggested by the software for methanolysis temperature, time, and mass ratio of MeOH to PET were 298.18 °C, 112 min and 6.01, respectively. Although the condition could not be obtained for actual experiment, an approximate optimum conditions (298 °C, 112 min, and 6) produced 99.79% at a high level of the DMT yield, well in agreement with the predicted value of 99.80%. The result was compared with those reported by other researchers for non-catalytic PET depolymerization in supercritical methanol shown in Table 6. The optimum yield obtained in this study is relatively higher than those reported in the literature.

In addition, compared with the previous work, RSM method is involved in the result analysis, which can propose the functional relationship between the DMT yield and control variables shown as Eq. (5) other than discrete optimal operation points in previous researches. In consequence, the optimal solution proposed by this study is more flexible and valuable than before when considering the varied real operating conditions in PET recycling industry.

4. Conclusion

Under the supercritical methanol, PET can be rapidly and completely degraded to monomer DMT. For analyzing and optimizing the yield of DMT in the PET depolymerization process, the quadratic equation with square root transformation modified fitted for DMT yield with the process variables was established by response surface methodology. ANOVA was also used to evaluate the significant terms and interaction effects and the validation of the model. Using RSM, we could found that the MeOH to PET mass ratio and relative term exhibit a less significant influence on the response surface in comparison to reaction temperature and time in the range of the experiment conditions. The optimum operating condition for methanolysis temperature, time, and mass ratio of MeOH to PET were 298 °C, 112 min and 6, respectively. Actual experiment based on the optimum conditions led to a high DMT yield of 99.79%, in well agreement with the predicted value.

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